

Perfect entropy functions of the Lattice Boltzmann method

To cite this article: I. V. Karlin *et al* 1999 *EPL* **47** 182

View the [article online](#) for updates and enhancements.

You may also like

- [Uniform Polynomial Equations Providing Higher-order Multi-dimensional Models in Lattice Boltzmann Theory](#)
Jae Wan Shim
- [Entropy and weak solutions in the lattice Bhatnagar—Gross—Krook model](#)
Zheng Ran, , Yu-Peng Xu et al.
- [On the Hydrodynamic Behaviour of the Lattice Boltzmann Equation](#)
M. Vergassola, R. Benzi and S. Succi

Perfect entropy functions of the Lattice Boltzmann method

I. V. KARLIN(*), A. FERRANTE and H. C. ÖTTINGER

*ETH Zürich, Department of Materials, Institute of Polymers
CH-8092 Zürich, Switzerland*

(received 4 January 1999; accepted in final form 19 May 1999)

PACS. 47.11+j – Computational methods in fluid dynamics.

Abstract. – In this letter, we derive entropy functions whose local equilibria are suitable to recover the Navier-Stokes equations in the framework of the Lattice Boltzmann method. For the two-dimensional nine-velocity lattice we demonstrate that such an entropy function is unique, and that the expansion of the corresponding local equilibrium is the well-known local equilibrium of Y. H. Qian *et al.* (*Europhys. Lett.*, **17** (1992) 479). Based on the knowledge of entropy functions, we introduce a new version of the Lattice Boltzmann method with an H -theorem built in.

Since the pioneering work on the Lattice Gas [1], the lattice-based approach to simulation of hydrodynamics received considerable attention over the past decade. In the widely used Lattice Boltzmann method [2], one considers populations of fictitious particles, $N_i(\mathbf{r}, t)$, where $i = 1, \dots, b$ labels discrete velocities $\mathbf{c}_i$. The set of discrete velocities, which can also include a zero vector (“rest population”), is associated with outgoing links at each site $\mathbf{r}$ of a regular isotropic lattice. Populations are updated at discrete time steps t according to an equation,

$$N_i(\mathbf{r} + \mathbf{c}_i, t + 1) - N_i(\mathbf{r}, t) = \Delta_i. \quad (1)$$

In the following, we restrict our attention to the isothermal Navier-Stokes equation. For that case, the collision integral Δ_i must obey only the local conservation laws $\sum_{i=1}^b \{1, c_{i\alpha}\} \Delta_i = 0$ for the local hydrodynamic fields, *i.e.*, the density $\rho = \sum_{i=1}^b N_i(\mathbf{r}, t)$, and momentum $\rho u_\alpha = \sum_{i=1}^b c_{i\alpha} N_i(\mathbf{r}, t)$. Here $\alpha = 1, \dots, d$ label the Cartesian components of d -dimensional vectors. If the long-time large-scale limit of eq. (1) recovers the Navier-Stokes equation, then hydrodynamics is implemented in a fairly simple, fully discrete kinetic picture. In the following, we denote $\mathbf{N}$ as the b -dimensional vector of populations.

An important part of any realization of the Lattice Boltzmann method is the problem of the local equilibrium $\mathbf{N}^{\text{eq}}$. From the perspective of classical kinetic theory, local equilibria are found as minimum points of a convex function $H(\mathbf{N})$, subject to constraints fixed by the hydrodynamic fields,

$$\sum_{i=1}^b \{1, c_{i\alpha}\} N_i^{\text{eq}} = \{\rho, \rho u_\alpha\}. \quad (2)$$

(*) Permanent address: Institute of Computational Modeling, Krasnoyarsk, Russian Federation.
E-mail: ikarlin@ifp.mat.ethz.ch

The convex function H (called the entropy function in the sequel) plays the role of the Boltzmann H -function. In addition, in order to recover the Navier-Stokes equation up to second-order accuracy in u , the local equilibrium must respect the condition

$$\sum_{i=1}^b c_{i\alpha} c_{i\beta} N_i^{\text{eq}} = \rho u_\alpha u_\beta + \rho c_s^2 \delta_{\alpha\beta}, \quad (3)$$

where c_s is the constant speed of sound. The principal question of the Lattice Boltzmann method is: *Do there exist entropy functions such that the corresponding local equilibria satisfy simultaneously the additional condition (3) to the order of accuracy of the method?* Following ref. [3], such entropy functions will be named “perfect”. The Boltzmann entropy function is perfect in the context of the classical continuous Boltzmann equation, which is the only kinetic theory known so far where the above question is answered. In the context of the Lattice Boltzmann method, this question has an important practical motivation concerning the stability of the practical realization. Indeed, if the classical definition of local equilibria is relaxed, it is possible to introduce a local equilibrium *ansatz* $\tilde{\mathbf{N}}^{\text{eq}}$ (typically, a second-order polynomial in u_α) which satisfies both conditions (2) and (3), disregarding any difference in their origin. The choice of the *ansatz* is usually not unique, and realizations may differ drastically: some are relatively stable while the others are not. On the other hand, if the local equilibrium is supported by *some* entropy function, then the Lattice Boltzmann method can be equipped with the H -theorem (first attempts in this direction for the Lattice BGK method have been done recently [4, 5]), and stability problem can be studied in a controlled way. However, within any approach which attempts to enhance stability via an H -theorem, only perfect entropy functions are relevant to recover the Navier-Stokes equations.

In this letter, we demonstrate that entropy functions which are perfect within the accuracy of the Lattice Boltzmann method do exist. For the sake of presentation, we first explain the method of their construction for a simple example. Next we study a particularly important two-dimensional nine-velocity lattice, and find that the perfect entropy function is unique and Boltzmann-like. Finally, based on the knowledge of the perfect entropy functions, we develop a realization of the Lattice Boltzmann method with the H -theorem built in.

The simplest example. – Let us first consider a one-dimensional lattice with spacing c . The velocity set at each lattice site consists of three velocities, $c_+ = c$, $c_- = -c$, and $c_0 = 0$. We consider H -functions of the form

$$H = h_0(N_0) + h_1(N_+) + h_1(N_-). \quad (4)$$

Here h_0 and h_1 are two convex functions which are yet unknown. The local equilibrium is found as the minimum of H , subject to given hydrodynamic fields ρ and ρu . Denoting $\mu_{0,1}(x) = [h'_{0,1}(x)]^{-1}$ the inverse of the derivatives of the functions $h_{0,1}(x)$, a formal result of this minimization reads: $N_0^{\text{eq}} = \mu_0(a)$, and $N_\pm^{\text{eq}} = \mu_1(a \pm \lambda c)$. The Lagrange multipliers a and λ are related to ρ and ρu by means of the constraints (2): $\mu_0(a) + \mu_1(a + \lambda c) + \mu_1(a - \lambda c) = \rho$, and $c\mu_1(a + \lambda c) - c\mu_1(a - \lambda c) = \rho u$. The additional condition (3) then reads: $c^2\mu_1(a + \lambda c) + c^2\mu_1(a - \lambda c) = \rho u^2 + \rho c_s^2$. For the time being, the sound speed c_s will be considered as a free parameter. Expressing the right-hand side of the latter equation in terms of μ_k with the help of the constraints, we find

$$\begin{aligned} T[\mu_0, \mu_1, c_s^2] &= c^2[\mu_1(a + \lambda c) + \mu_1(a - \lambda c)] - \frac{c^2[\mu_1(a + \lambda c) - \mu_1(a - \lambda c)]^2}{\mu_0(a) + \mu_1(a + \lambda c) + \mu_1(a - \lambda c)} - \\ &- c_s^2[\mu_0(a) + \mu_1(a + \lambda c) + \mu_1(a - \lambda c)] = 0. \end{aligned} \quad (5)$$

Equation (5) is a nonlinear functional equation for the functions $\mu_{0,1}$. We shall now solve it approximately by using a Taylor expansion to order λ^2 : $\mu_1(a \pm \lambda c) = \mu_1(a) \pm \mu'_1(a)\lambda c + (1/2)\mu''_1(a)(\lambda c)^2 + \dots$, where primes denote derivatives in the point a . The status of this expansion will be discussed below. Substituting this expansion into eq. (5), we require that terms of the order λ^0 and λ^2 are equal to zero (terms of the order λ cancel out identically). After a few algebra one obtains two equations:

$$\mu_0 = 2 [(c/c_s)^2 - 1] \mu_1, \quad \mu'_1 \mu'_1 = (1/2) [(c/c_s)^2 - 1] \mu_1 \mu''_1. \quad (6)$$

The parameter c_s must now be chosen in such a way that the differential equation in the last line of eq. (6) admits solutions compatible with the convexity requirement for the H -function (4). In particular, for $c_s^2 = (1/3)c^2$, the resulting differential equation, $\mu_1 \mu''_1 = \mu'_1 \mu'_1$, has a solution represented by $\mu_1(a) = \exp[a]$, and it follows that $\mu_0(a) = 4 \exp[a]$. This means that functions of the form $\mu_1(a \pm \lambda c) = \exp[a \pm \lambda c]$, and $\mu_0(a) = 4 \exp[a]$, satisfy eq. (5) to order λ^4 , for arbitrary a . The latter statement can be written as: $T[4 \exp[a], \exp[a \pm \lambda c], c^2/3] = \varphi(a)(\lambda c)^4 + \dots$, where φ is some regular function. Finally, the solution found is the local equilibrium of the convex Boltzmann-like H -function:

$$H = N_0(\ln N_0 - 1 - \ln 4) + N_+(\ln N_+ - 1) + N_-(\ln N_- - 1). \quad (7)$$

Thus, the entropy function (7) is perfect to the order λ^4 . Note the different role of the rest population with respect of moving particles in this expression. Several remarks are now in order:

i) *Accuracy.* As is well known, the Lagrange multipliers behave for small u as follows: $a = a_0 + u^2 a_1 + \dots$, and $\lambda = u \lambda_1 + \dots$, where $a_{0,1}$ and λ_1 are constants. Thus, $T \sim O(u^4)$, and our local equilibrium satisfies condition (3) not exactly but within the overall accuracy of the Lattice Boltzmann method.

ii) *Expansion in Lagrange multipliers.* This expansion should not be confused with the well-known expansion in terms of the average velocity. The coefficients in the former are functions of the Lagrange multiplier a , which in turn is a function of u ; results are derived by solving differential equations as in (6), whereas the expansion in terms of the average velocity leads to a purely algebraic problem.

iii) *Other solutions and local equilibrium ansatz.* A different choice of the sound speed may result in different entropy functions. For instance, $c_s^2 = c^2/5$ in eq. (6) leads to $\mu_1 = a^2$, $\mu_0 = 8a^2$, which corresponds to the perfect H -function $H = (2/3)[N_+^{3/2} + N_-^{3/2} + (1/\sqrt{8})N_0^{3/2}]$. In this case, the local equilibrium is an explicit function of the hydrodynamic fields [4], and we shall use this solution here as an example for demonstrating the link of our approach with other methods. We have: $N_\pm^{\text{eq}} = (\rho/20)[R \pm (u/2c) + R^{-1}(u/2c_s)^2]$, $N_0^{\text{eq}} = (2\rho/5)R$, where $R = 1 + [1 - (u/c_s)^2]^{1/2}$. When these functions are expanded up to terms of order u^2 , and when the higher-order terms are *neglected*, the result is a polynomial quadratic in u . This polynomial satisfies condition (3) *exactly*, and provides the quadratic local equilibrium ansatz. This remark is quite general: *the second-order expansion of the local equilibrium corresponding to a perfect entropy function is itself the local equilibrium ansatz which satisfies condition (3) exactly.* As long as the deviation from equilibrium is small, and populations stay positive, the local equilibrium ansatz [6] minimizes the entropy function (7).

iv) *Energy shells and the number of constraints.* In choosing the initial form of the H function, eq. (4), we have followed an idea of organizing the velocity set into energy shells E_k , each shell being the subset of velocities with equal magnitude. Each shell k is characterized by its individual function h_k in the expression (4), and it is possible to satisfy the two equations (6)

since we have sufficient “degrees of freedom” in the problem (two functions, h_0 and h_1 , and one parameter c_s). The idea of energy shells is borrowed from ref. [7].

Two-dimensional case. – To construct H -functions in higher dimensions, it is desirable to have at least as many energy shells as the number of equations arising in the second-order expansion of the functional equations like eq. (5). In this respect, an interesting example is provided by the well-known two-dimensional nine-velocity lattice (2d9v). In a two-dimensional Cartesian coordinates system, the set of velocities is organized in three energy shells: $E_0 = \{(0, 0)\}$ (population N_0), $E_1 = \{(c, 0), (0, c), (-c, 0), (0, -c)\}$ (populations N_{11} , N_{12} , N_{13} and N_{14}), and $E_2 = \{(c, c), (-c, c), (-c, -c), (c, -c)\}$ (populations N_{21} , N_{22} , N_{23} and N_{24}). This allows us to introduce three functions h_0 , h_1 , and h_2 , and to write the desired H -function as $H = h_0(N_0) + \sum_l h_1(N_{1l}) + \sum_l h_2(N_{2l})$. Minimization of this function, subject to the hydrodynamic constraints, gives three equations, relating the functions $\mu_k(a + \lambda_\beta c_{i\beta})$ to the hydrodynamic fields ρ and ρu_α . Proceeding as before, we express the right-hand side of eq. (3) in terms of the functions μ_k , using the hydrodynamic constraints, and expand the functions μ_k in the point a up to second-order terms in $\lambda_\alpha c_{i\alpha}$. With this, we derive *four* equations depending on the parameter c_s for only *three* functions μ_k to cancel terms of order 1, λ_x^2 , λ_y^2 , and $\lambda_x \lambda_y$. This system is compactly written in terms of functions, $Z = \mu_0 + 4(\mu_1 + \mu_2)$, $R = \mu_1 + 2\mu_2$, and μ_2 :

$$Z = 2(c/c_s)^2 R, \quad \mu_2'' = (1/2)(c/c_s)^2 R'', \quad R' R' = R R'', \quad R' R' = (1/2)[(c/c_s)^2 - 1] R R''. \quad (8)$$

The latter system is consistent *if and only if* $c_s^2 = c^2/3$. In this case we have the exponential solution, and recover the unique perfect Boltzmann-like entropy function:

$$H = H_B + \ln(3/8)N_0 + \ln(3/2) \sum_{l=1}^4 N_{1l} + \ln(6) \sum_{l=1}^4 N_{2l}, \quad (9)$$

where $H_B = \sum_{k=1}^2 \sum_{l=1}^4 N_{kl}(\ln N_{kl} - 1)$. Notice that convexity is a stringent requirement on admissible solutions of equations like (6) or (8), and neither the existence of such a solution nor its Boltzmann-like form were assumed *a priori*. The local equilibrium corresponding to the entropy function (9) is: $N_0^{\text{eq}} = (8/3)\exp[a]$, $N_{1l}^{\text{eq}} = (2/3)\exp[a + \lambda_\alpha c_{1l\alpha}]$, and $N_{2l}^{\text{eq}} = (1/6)\exp[a + \lambda_\alpha c_{2l\alpha}]$. Expanding the Lagrange multipliers in terms of u_α , we find $a = \ln(\rho/6) - [u^2/(2c_s^2)]$ and $\lambda_\alpha = u_\alpha/c_s^2$, for small u . Using this result, we find that the corresponding local equilibrium ansatz is identical to the well-known ansatz of Y. H. Qian, D. d’Humières and P. Lallemand [7]. The latter identification explains *why* this ansatz is more stable than any other quadratic ansatz on the 2d9v lattice: to the order of approximation, it is supported by the entropy function (9).

Stabilization procedure. – Knowledge of the perfect-entropy function suggests realizations of the Lattice Boltzmann method which do not require an explicit expression for the local equilibrium. Below, we shall formulate one of such realizations. Before doing this, however, we shall describe a general procedure which equips *any* realization with the H -theorem. For the time being, let us assume that the collision integral $\Delta(\mathbf{N})$ in the kinetic equation (1) is realized in such a way that it satisfies two conditions: i) the conservation laws $((\Delta, \{\mathbf{1}, \mathbf{c}_\alpha\}) = 0$, where $\mathbf{1} = (1, \dots, 1)$ and $\mathbf{c}_\alpha = (c_{1\alpha}, \dots, c_{b\alpha})$, and $(\cdot, \cdot)$ is the standard scalar product of b -dimensional vectors), and ii) the entropy production inequality, $(\nabla H, \Delta) \leq 0$, where ∇H is the gradient of the entropy function, while the equality sign implies $\mathbf{N} = \mathbf{N}^{\text{eq}}$. A collision integral which meets these requirements will be termed admissible. For instance, the standard BGK collision integral is admissible. For each pair of vectors $\{\mathbf{N}, \Delta\}$, where $\mathbf{N}$ is a population vector, $N_i \geq 0$,

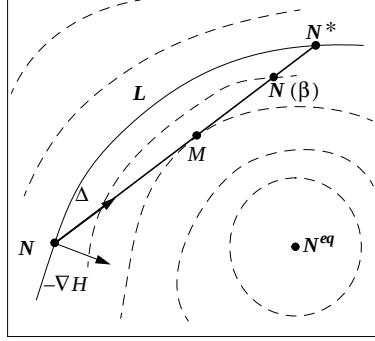


Fig. 1. – Stabilization procedure, bulk case. The curves represent entropy levels, surrounding the local equilibrium $\mathbf{N}^{\text{eq}}$. The solid curve L is the entropy level with the value $H(\mathbf{N}) = H(\mathbf{N}^*)$, where $\mathbf{N}$ is the initial, and $\mathbf{N}^*$ is the auxiliary population. The vector Δ represents the collision integral, the sharp angle between Δ and the vector $-\nabla H$ reflects the entropy production inequality. The point $\mathbf{M}$ is the minimum entropy state on the segment $[\mathbf{N}, \mathbf{N}^*]$ (see also ref. [8]). The result of the collision update is represented by the point $\mathbf{N}(\beta)$. The choice of β shown corresponds to the “overrelaxation”: $H(\mathbf{N}(\beta)) > H(\mathbf{M})$ but $H(\mathbf{N}(\beta)) < H(\mathbf{N})$. The particular case of the BGK collision (not shown) would be represented by a vector Δ_{BGK} , pointing from $\mathbf{N}$ towards $\mathbf{N}^{\text{eq}}$, in which case $\mathbf{M} = \mathbf{N}^{\text{eq}}$.

and Δ is the corresponding value of the collision integral, we introduce an *auxiliary population vector* $\mathbf{N}^* = \mathbf{N} + \alpha^* \Delta$. The scalar parameter α^* is derived as follows: Let us consider the equation

$$H(\mathbf{N}) = H(\mathbf{N} + \alpha \Delta). \quad (10)$$

There are two cases classified by the number of solutions eq. (10) may have. In the first case, eq. (10) has two solutions, $\alpha_1 = 0$, and another solution α_2 (notice that the degeneracy $\alpha_2 = \alpha_1 = 0$ occurs only if $\mathbf{N} = \mathbf{N}^{\text{eq}}$). In this case, the parameter α^* of the auxiliary population is taken as $\alpha^* = \alpha_2$. This situation can be interpreted as the “bulk case” since both vectors, $\mathbf{N}$ and $\mathbf{N}^*$, are located in the interior of the phase space of populations. The second (“boundary”) case corresponds to the situation when eq. (10) has only one (nondegenerate) solution $\alpha_1 = 0$. Then $|\alpha^*| = \min_{i=1, \dots, b; \Delta_i \neq 0} \{N_i / |\Delta_i|\}$, $\text{sgn} \alpha^* = -\text{sgn} \Delta_{i^*}$, where i^* is the component which realizes the minimum: The auxiliary state $\mathbf{N}^*$ is taken at the boundary of the phase space (at least one of the populations N_i^* is equal to zero).

The auxiliary populations have a simple geometrical interpretation: Consider the hypersurface of constant entropy L which includes the point $\mathbf{N}$. The auxiliary state $\mathbf{N}^*$ is located on a line emerging from the point $\mathbf{N}$ in direction Δ . This line can cross L once more, then $\mathbf{N}^*$ is defined as the second point of crossing (convexity of entropy prohibits more than two crossings), and this corresponds to the bulk case mentioned above. The boundary case corresponds to the situation when the line leaves the phase space before it crosses L for the second time. Then the state $\mathbf{N}^*$ is taken as the point where the line crosses the boundary of the phase space. The bulk case is visualized in fig. 1.

The auxiliary population sets the limit of the collision update in such a way that the entropy function H decreases in the result. Once the auxiliary population $\mathbf{N}^*$ is defined, the result of the collision is set as $\mathbf{N}(\beta) = (1 - \beta)\mathbf{N} + \beta\mathbf{N}^*$, where β is a fixed parameter chosen on the segment $[0, 1]$. Convexity of the entropy function implies the following inequality (*the local H-theorem*): $H(\mathbf{N}(\beta)) \leq H(\mathbf{N})$. Moreover, when approaching the hydrodynamic regime, *i.e.* close enough to the local equilibrium, only the bulk case is realized because $\mathbf{N}^{\text{eq}}$ for

Boltzmann-like entropy functions is a positive vector. Then the parameter β controls the viscosity coefficient in the resulting Navier-Stokes equations in the following way: The zero viscosity limit corresponds to $\beta \rightarrow 1$. For the BGK collision integral, this statement follows from the proof of the H -theorem [4], and it can be extended to other realizations.

Realization. – Finally, we present an explicit realization of the collision integral based on the knowledge of the entropy. Let $\mathbf{g}_{(s)}$, $s = 1, \dots, b - (d + 1)$, be a basis (not obligatory orthonormal) of the subspace orthogonal to vectors of conservation laws $\mathbf{1}$ and $\mathbf{c}_\alpha$. For each vector $\mathbf{g}_{(s)}$, we define a decomposition $\mathbf{g}_{(s)} = \mathbf{g}_{(s)}^+ - \mathbf{g}_{(s)}^-$, where all components of vectors $\mathbf{g}_{(s)}^\pm$ are nonnegative, and if $g_{(s)i}^\pm \neq 0$, then $g_{(s)i}^\mp = 0$. Let us consider the collision integral of the form

$$\Delta = \sum_{(s)} \gamma_{(s)} \mathbf{g}_{(s)} \left\{ \exp \left[\left(\nabla H, \mathbf{g}_{(s)}^- \right) \right] - \exp \left[\left(\nabla H, \mathbf{g}_{(s)}^+ \right) \right] \right\}. \quad (11)$$

Here $\gamma_{(s)} > 0$. By the construction, the collision integral (11) is admissible. If the entropy function is Boltzmann-like, and the components of vectors $\mathbf{g}_{(s)}$ are integers, the collision integral assumes the familiar Boltzmann-like form. Equation (11) is motivated by the models developed in [9]. For the entropy function (7), the resulting kinetic equation is identical to the model studied in [6]. Here we shall specify the collision integral (11) for the 2d9v lattice with the entropy (9).

The population vector is $\mathbf{N} = (N_0, N_{11}, N_{12}, N_{13}, N_{14}, N_{21}, N_{22}, N_{23}, N_{24})$. A suitable basis $\mathbf{g}_{(s)}$ is provided by the well-known collisional basis [10]: $\mathbf{g}_{(1)} = 6\mathbf{e}_1$, $\mathbf{g}_{(2)} = 2\sqrt{3}\mathbf{e}_4$, $\mathbf{g}_{(3)} = 2\sqrt{3}\mathbf{e}_5$, $\mathbf{g}_{(4)} = 2\mathbf{e}_6$, $\mathbf{g}_{(5)} = 2\mathbf{e}_7$, and $\mathbf{g}_{(6)} = 6\mathbf{e}_8$, where $\mathbf{e}_i$ are orthonormal vectors given by eq. (11) of ref. [10], while factors are introduced in order to make all components integer. The decomposition $\mathbf{g}_{(s)}^\pm$ is straightforward: For instance, for $\mathbf{g}_{(6)} = (4, -2, -2, -2, -2, 1, 1, 1, 1)$, we have $\mathbf{g}_{(6)}^+ = (4, 0, 0, 0, 0, 1, 1, 1, 1)$ and $\mathbf{g}_{(6)}^- = (0, 2, 2, 2, 2, 0, 0, 0, 0)$, and so forth. Derivation of the collision integral (11) for the entropy (9) amounts to a simple algebra with the result

$$\begin{aligned} \Delta_0 &= -4R_1^{(8)} - 4R_2^{(8)}, \\ \Delta_{11} &= -R_1^{(2)} - 2R_1^{(4)} - R_1^{(8)} + 2R_2^{(8)}, & \Delta_{21} &= -R_2^{(2)} + R_1^{(4)} + R_2^{(4)} + 2R_1^{(8)} - R_2^{(8)}, \\ \Delta_{12} &= R_1^{(2)} - 2R_2^{(4)} - R_1^{(8)} + 2R_2^{(8)}, & \Delta_{22} &= R_2^{(2)} - R_1^{(4)} + R_2^{(4)} + 2R_1^{(8)} - R_2^{(8)}, \\ \Delta_{13} &= -R_1^{(2)} + 2R_1^{(4)} - R_1^{(8)} + 2R_2^{(8)}, & \Delta_{23} &= -R_2^{(2)} - R_1^{(4)} - R_2^{(4)} + 2R_1^{(8)} - R_2^{(8)}, \\ \Delta_{14} &= R_1^{(2)} + 2R_2^{(4)} - R_1^{(8)} + 2R_2^{(8)}, & \Delta_{24} &= R_2^{(2)} + R_1^{(4)} - R_2^{(4)} + 2R_1^{(8)} - R_2^{(8)}. \end{aligned} \quad (12)$$

Here $R_1^{(2)} = N_{11}N_{13} - N_{12}N_{14}$, $R_2^{(2)} = N_{21}N_{23} - N_{22}N_{24}$, $R_1^{(4)} = N_{11}^2N_{22}N_{23} - N_{13}^2N_{21}N_{24}$, $R_2^{(4)} = N_{12}^2N_{23}N_{24} - N_{14}^2N_{21}N_{22}$, $R_2^{(8)} = N_0^4N_{21}N_{22}N_{23}N_{24} - N_{11}^2N_{12}^2N_{13}^2N_{14}^2$, and

$$R_1^{(8)} = \theta_1^{(8)} N_0^4 N_{11} N_{12} N_{13} N_{14} - N_{21}^2 N_{22}^2 N_{23}^2 N_{24}^2, \quad \theta_1^{(8)} = (1/2)^{24}. \quad (13)$$

Equation (12) is a representative of the six-parametric family of admissible collision integrals, other representatives are obtained by multiplying polynomials $R_i^{(j)}$ with positive constants $\gamma_i^{(j)}$.

Polynomials $R_i^{(j)}$ are naturally interpreted in terms of the elementary collision processes, where the degree of the polynomial j indicates the number of participating particles. It is crucial to notice that the use of the perfect entropy (9) amounts to a *large disparity* between the rates of the direct and the inverse collisions in the eight-particle process (13) imprinted by the factor $\theta_1^{(8)}$. The latter is directly related to the logarithmic factors in eq. (9): In the *gain* term of eq. (13), each of the four rest particles contributes the factor $1/8$, while each of the

four particles of the shell E_1 contribute $1/2$, thus, in total, the gain term receives the factor $\theta_1^{(8)+} = (1/2)^{16}$. On the other hand, in the *loss* term, each of the eight particles of the shell E_2 contributes the factor 2, and thus $\theta_1^{(8)-} = 2^8$. Since the total rate (“gain minus loss”) can be multiplied with arbitrary factor, only the ratio $\theta_1^{(8)} = \theta_1^{(8)+}/\theta_1^{(8)-}$ is relevant, and it is therefore *the constant* characterizing the corresponding elementary collision process. A similar consideration shows that the ratios $\theta_i^{(j)}$ are equal to one for both the two-particle, for both the four-particle, and for the remaining eight-particle processes, and they are omitted in the above expressions. If the Boltzmann entropy were used instead of eq. (9), the result would be trivial: $\theta_i^{(j)} = 1$ for all the six elementary processes.

Preliminary numerical tests were focused, in the first place, on the accuracy of the approximately perfect entropy functions. In the $1d$ example (7), the effective sound speed $\tilde{c}_s^2[u] = \sum_i N_i^{\text{eq}}[u] c_i^2 - u^2$ was considered as a function of the average velocity u ($\rho = 1$). If the perfect entropy (7) were an exact solution, we would have $c_s^2 = (1/3)c^2$ for all u . We have found that though $\tilde{c}_s^2$ is a decreasing function of $|u|$, its deviation from $c_s^2 = (1/3)c^2$ does not exceed 5% at Mach number one. This indicates that approximately perfect entropies should not bring large numerical discrepancies in the stress tensor in the domain of validity. The second test concerns implementation of the stabilization procedure which requires solution of eq. (10) at each time step for each lattice site (notice that eq. (10) is one-dimensional regardless of the choice of the lattice). Application of the Newton method demonstrates that several iterations (≤ 8) are needed to find the auxiliary state, their number usually increases when the initial state is far from equilibrium. Taking into account the fact that costs for computation of the collision integral (11) are approximately the same as in the usual schemes, the fast solvers for eq. (10) is key to implementation. A detailed study of the model (12) will be reported elsewhere.

Conclusions. – Upon the derivation of the perfect entropy functions of the isothermal Lattice Boltzmann method, we have developed two applications: the nonlinear stabilization procedure, and the general class of admissible Lattice Boltzmann kinetic equations.

IVK acknowledges discussion of results with B. CHOPARD, A. NAKKASYAN and S. SUCCI.

REFERENCES

- [1] FRISCH U., HASSLACHER B. and POMEAU Y., *Phys. Rev. Lett.*, **56** (1986) 1505.
- [2] For a review see QIAN Y. H., SUCCI S. and ORSZAG S., *Annu. Rev. Comput. Phys.*, **3** (1995) 195.
- [3] KARLIN I. V. and SUCCI S., *Phys. Rev. E*, **58** (1998) R4053.
- [4] KARLIN I. V., GORBAN A. N., SUCCI S. and BOFFI V., *Phys. Rev. Lett.*, **81** (1998) 6.
- [5] WAGNER A. J., *Europhys. Lett.*, **44** (1998) 144.
- [6] QIAN Y. H., D’HUMIÈRES D. and LALLEMAND P., in *Advances in Kinetic Theory and Continuum Mechanics*, edited by R. GATIGNOL and SOUBBARAMAYER (Springer, Berlin) 1991, p. 127.
- [7] QIAN Y. H., D’HUMIÈRES D. and LALLEMAND P., *Europhys. Lett.*, **17** (1992) 479.
- [8] Constructive estimations of the state $\mathbf{M}$ for a generic kinetic equation are given in: GORBAN A. N., KARLIN I. V., ZMIEVSKII V. B. and NONNENMACHER T., *Physica A*, **231** (1996) 648.
- [9] DUKEK G., KARLIN I. V. and NONNENMACHER T., *Physica A*, **239** (1997) 493 and references therein.
- [10] D’HUMIÈRES D., *AIAA Rarefied Gas Dynamics: Theory and Applications*, Vol. **159** of *Progress in Astronautics and Aeronautics* (1992) 450.